

NIDHI HARISHCHANDRA FALAK

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EDUCATION

University of South Florida , Tampa, FL, USA	
Master of Science in Business Analytics and Information Systems	May 2025
Courses: Data Mining, Analytical Methods of Business, Advanced Database Management, Data Visualization, Big Data, Cloud Solutions Architecture, Data Warehousing, Distributed Information Systems	GPA: 3.9/4
University of Mumbai , Mumbai, India	June 2021
Bachelor of Engineering in Electronics	GPA: 7.7/10

TECHNICAL SKILLS

Programming Languages:	Python, R, SQL, HTML, CSS, JavaScript.
Databases:	MySQL, Oracle, T-SQL, PL/SQL, SQLite, PostgreSQL, Cassandra, Snowflake, Redshift, MongoDB
Business Intelligence Tools:	Power BI, Tableau, Qlik, Looker Studio, SSRS, SSIS, Power Query, DAX
Frameworks, Tools & Libraries:	TensorFlow, PySpark, Hadoop, Microsoft Excel, JIRA, Confluence, SSMS, Databricks, Git, Docker, Salesforce, ServiceNow, Siebel, Dynatrace, Visual Studio Code, NumPy, Pandas, Scikit-learn, Seaborn, Apache Spark.
Certifications:	Microsoft PL-300: Power BI Data Analyst, AWS Certified Cloud Practitioner

PROFESSIONAL WORK EXPERIENCE

University of South Florida, USA Data Analytics Research Assistant	October 2024 – May 2025
- Created and implemented machine learning models (XG Boost, Random Forest & Prophet) to forecast regional trends for the Tampa Bay State of the Region Report, achieving 90% accuracy and providing valuable insights for policymaker decision-making across 20 MSAs. - Integrated Principal Component Analysis (PCA) with predictive models to compare economic, workforce, and education metrics, increasing predictive accuracy by 20% and enabling more effective regional planning and data-driven decision-making - Designed and launched interactive Tableau dashboards to track KPIs across multiple sectors, driving actionable decisions and helping local businesses save up to \$80k annually through operational optimization.	
University of South Florida, USA Data Analyst	April 2024 – September 2024
- Analyzed library inventory and procurement data using DAX and Power Query to identify trends in book demand, supplier lead times, and usage patterns, improving resource allocation by 20%. - Automated Power BI dashboards with SQL-based pipelines to track inventory turnover, reorder levels, and acquisition cycles, reducing manual reporting by 40% and enhancing real-time visibility. - Built demand forecasting models in Python using Pandas and Scikit-learn to optimize book procurement and stock levels, resulting in a 30% reduction in shortages and a 15% increase in circulation efficiency.	
Tata Consultancy Services, India Assistant System Engineer	July 2021 – July 2023
- Engineered and optimized ETL pipelines for financial data integration, automating loan processing and risk analysis using SSIS and AWS (EC2, RDS, Redshift) at BMO Bank, US, reducing processing time by 30% and improving loan approval accuracy by 25%. - Built and maintained 50+ Power BI and Tableau dashboards, providing actionable insights into loan performance, asset utilization, and portfolio risk, driving \$2M in revenue growth through effective data storytelling. - Created SSRS reports for real-time financial tracking, increasing visibility into loan arrears, asset depreciation, and credit risk, reducing overdue loans by 25% and streamlining financial decision-making. - Collaborated with business stakeholders in an Agile environment to design user-centric reports, enhancing risk assessment, credit decisioning, and financial forecasting, leading to a 15% improvement in decision making across banking, investment, and transportation finance sectors.	

PROJECT EXPERIENCE

Predicting Churn Customers
- Analyzed 100,000+ customer profiles using Pandas, performing data cleaning and feature engineering (e.g., transaction averages, days since last purchase) to identify churn patterns and drive actionable insights for retention strategies. - Deployed an XG Boost model with 96% accuracy to predict customer churn, compared with other models (Random Forest, SVM, Decision Tree), and integrated predictions into the CRM for real-time personalized marketing, reducing churn by 20% and boosting Customer Lifetime Value (CLV) by 20%, driving revenue growth.
Healthcare Analytics Dashboard
- Developed and optimized Power BI dashboards for patient flow monitoring, using advanced DAX measures to track key metrics, resulting in a 40% improvement in operational efficiency and a 30% increase in patient satisfaction. - Investigated large datasets to identify bottlenecks and inefficiencies, reducing wait times by 20% and improving scheduling accuracy, while generating KPI-driven reports that enhanced service delivery and overall care quality.
Racial Biases in Police Stops
- Analyzed 3M+ traffic stops data using Apache Spark and Hadoop for distributed processing, performing data cleaning, feature engineering, and EDA to uncover trends like racial disparities. - Developed machine learning models (Logistic Regression, Decision Tree, Linear SVC, MLP Classifier) in Spark ML, achieving 87% accuracy, and provided insights for policy interventions to improve equitable law enforcement.